

Model Question Paper

BIG DATA ANALYTICS

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions choosing at least **ONE** question from each **MODULE**

Module -1			*Bloom's Taxonomy Level	COs	Marks
Q.01	a	What is Big Data? Explain the challenges With Big Data.	L2	CO1	10
	b	What is big data analytics? Explain classification of analytics	L2	CO1	10
OR					
Q.02	a	Explain CAP Theorem.	L2	CO1	05
	b	What is NewSQL? Explain the Characteristics of NewSQL	L2	CO1	07
	c	Explain the Hadoop Ecosystem Components for Data Processing and Data Analysis	L2	CO1	08
Module-2					
Q. 03	a	With a neat diagram explain HDFS architecture.	L2	CO1	08
	b	Explain HDFS Daemons	L2	CO1	08
	c	Give HDFS Commands to perform the following operations i) To get the list of directories and files at the root of HDFS ii) To create a directory (say, sample) in HDFS iii) To copy a file from local file system to HDFS iv) To display the contents of an HDFS file on console	L3	CO2	04
OR					
Q.04	a	Explain anatomy of HDFS file read and write.	L2	CO1	10
	b	Implement a word count program in Hadoop.	L3	CO2	10
Module-3					
Q. 05	a	What is MongoDB? Why MongoDB? Illustrate the process of Creating or Generating a Unique Key.	L2	CO1	10
	b	Implement Functions: Count – Sort – Limit – Skip – Aggregate using MongoDB.	L3	CO1	10
OR					
Q. 06	a	Explain the following methods with an example for each i) Save() ii) Find() iii) Pretty() iv) Count() v)Skip()	L2	CO1	10
	b	Given a collection named Students with the following fields: StudRollNo, StudName, Grade, Hobbies, DOJ Write MongoDB commands to perform the following operations i) Insert the details of one student ii) Display the details of the student having rollno 101 iii) Change the hobby of the student named "Rahul" from Cricket to Football iv) Delete the record of the students belonging to V grade. v) Retrieve the details of students whose names start with the letter 'B'	L3	CO1	10

Module-4					
Q. 07	a	What is Hive? List the features of Hive.	L2	CO3	4
	b	Explain Hive File Formats	L2	CO3	8
	c	Explain bucketing with an example.	L2	CO3	8
OR					
Q. 08	a	What is Pig? explain the key features of Pig.	L2	CO3	4
	b	With a neat diagram explain the anatomy of pig	L2	CO3	8
	c	Explain any five relational operators of pig with an example for each.	L2	CO3	8
Module-5					
Q. 09	a	With a neat diagram explain the main feature of spark.	L2	CO3	10
	b	Explain the five layer architecture for running applications using spark stack.	L2	CO3	10
OR					
Q. 10	a	With a neat diagram explain text mining process.	L2	CO3	10
	b	Explain the page rank algorithm using the relative authority of the parents over linked children.	L2	CO3	10

